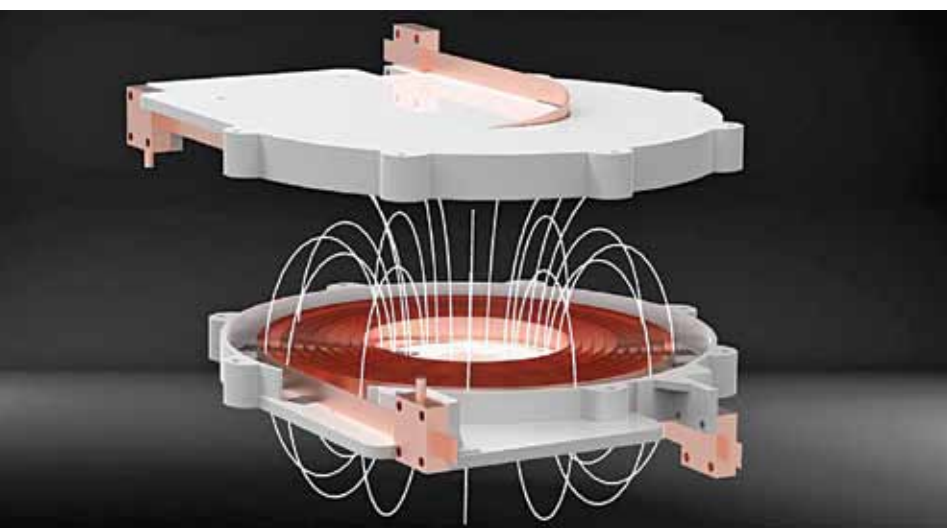




Contactless superconducting coils for better power transmission

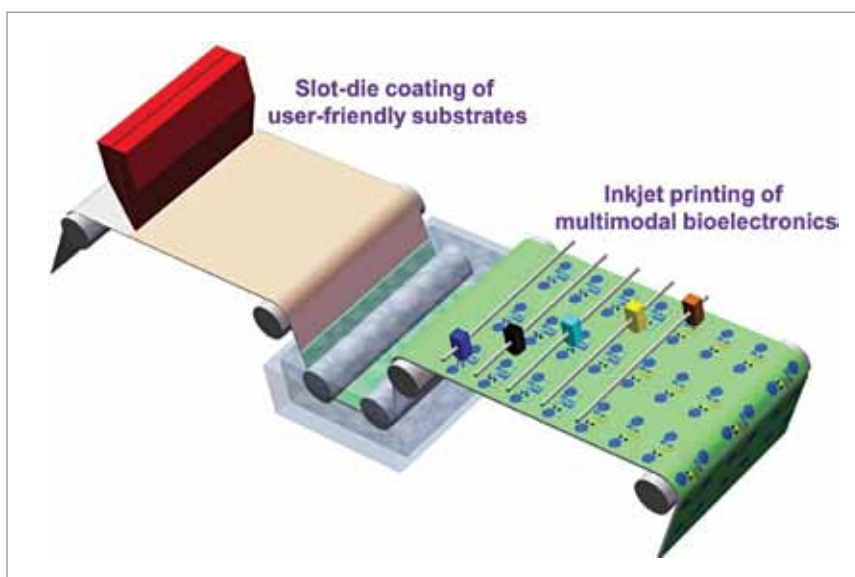


A coil made of superconducting wires that can contactlessly transmit power of more than five kilowatts without major losses (Credit: www.tum.de)

Contactless power transmission has established itself as a key technology when it comes to charging small electronic devices. However, the current transmission systems for high-performance recharging in the kilowatt range and above are large and heavy, since they are based on copper coils. Now a team of physicists at the Technical University of Munich (TUM) has succeeded in making a coil with superconducting wires that is capable of transmitting contactless power in the range of more than five kilowatts and with only small losses. Possible wide-scale applications include autonomous industrial robots, medical equipment, electric vehicles, as well as autonomous electric aircraft.

Covid-19 detection using bioelectronic substrate on skin

The demand for bioelectronic wearables has grown in recent years, largely fuelled by their ability to record and monitor a person's health—from heart rate to sleep quality. Soon, such bioelectronic devices would be able to wirelessly transmit a person's vital signs, providing early detection of health issues such as Covid-19 or heart disease to eliminate the need for in-person visits. With this in consideration, large-scale development of bioelectronic wearables that have breathable, waterproof, and passive cooling capabilities is set to take place at the College of Engineering, University of Missouri. And in order to mass-produce the bioelectronic sensors, these will be directly printed onto the supportive material using a method called mask-free inkjet printing.



A customisable device capable of simultaneously tracking multiple vital signs such as blood pressure, heart activity, and skin hydration (Credit: <https://engineering.missouri.edu>)